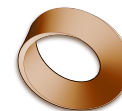


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics



1.5 – Longitudinal Waves

Unit 1: Oscillations & Waves

Learning intentions

- a) I can describe longitudinal waves as having oscillations parallel to the direction of energy transfer.
- b) I can identify **compressions** and **rarefactions** in diagrams and physical models of longitudinal waves.
- c) I can measure wavelength from one compression to the next adjacent compression.
- d) I can distinguish longitudinal waves from transverse waves based on particle motion relative to wave propagation.

Theory

In a **longitudinal wave**, the particles of the medium vibrate back and forth in the same direction that the wave travels. This is different from a transverse wave, where particles move perpendicular (at right angles) to the direction of travel. A common model for visualising longitudinal waves is a **slinky** spring. When you push and pull one end of a slinky along its length, you create regions where the coils are squashed together and regions where they are spread apart. These regions travel along the slinky, carrying energy from one end to the other.

The region where the particles are pushed close together is called a **compression**. In this area, the density of the medium is higher than normal. Conversely, the region where the particles are spread further apart is called a **rarefaction**. Here, the density is lower than normal. As the wave moves, compressions and rarefactions alternate. For example, when you speak, your vocal cords vibrate, creating areas of high pressure (compressions) and low pressure (rarefactions) in the air. These pressure variations travel to your friend's ear as a **sound** wave, which is the most common example of a longitudinal wave in everyday life.

To measure the **wavelength** (λ) of a longitudinal wave, you must measure the distance from the centre of one compression to the centre of the next adjacent compression. You can also measure from the centre of one rarefaction to the centre of the next rarefaction. It is important not to measure from a compression to a rarefaction, as that distance represents half a wavelength. The relationship between wave speed (v), frequency (f), and wavelength is given by the equation:

wave speed = frequency $\times$ wavelength, or $v = f\lambda$.

Consider a sound wave travelling through air with a frequency of 500 Hz. If the speed of sound in air is 340 m/s, we can calculate the wavelength. We rearrange the formula to make wavelength the subject: wavelength = wave speed / frequency. So, $\lambda = 340/500$. This gives $\lambda = 0.68$ m. Therefore, the distance between two consecutive compressions is 0.68 metres.

When looking at a diagram, you can identify a longitudinal wave by observing the particle spacing. If the diagram shows coils or particles bunched up in some areas and spread out in others, and an arrow indicates the wave moving horizontally along that same line, it is longitudinal. In contrast, a transverse wave diagram would show peaks (crests) and troughs, with particles moving up and down while the wave moves sideways.

Complete the sentence: In a longitudinal wave, the oscillations are _____ to the direction of energy transfer.

Word bank: perpendicular · parallel · compressions · rarefactions · wavelength

Activity

Slinky Compression Explorer

You will need a long slinky spring (at least 3 metres) and a partner. Work in pairs. One person holds one end of the slinky fixed against a wall or floor, while the other person holds the loose end. Ensure the slinky is stretched out but not pulled too tight.

- The holder at the fixed end keeps their hand steady.
- The second person quickly pushes the slinky forward towards the wall and then pulls it back sharply to create a pulse. This creates a compression that travels down the spring.
- Observe how the coils bunch up (compression) and spread out (rarefaction) as they move away from you.
- Now, try to create a continuous wave by pushing and pulling your end rhythmically. Keep the motion strictly along the length of the slinky.
- Ask your partner to mark two points on the floor: one where a compression is currently located, and another where the next compression will be when it reaches them. Measure the distance between these marks with a tape measure. This distance is the wavelength of your generated wave.
- Repeat step 5 three times, varying how fast you shake the slinky (frequency). Record your observations in the table below.

Trial	Description of motion (fast/slow)	Approximate wavelength observed (m)	Did comp
1		_____	
2		_____	
3		_____	

Questions

1. Define a compression in terms of particle spacing.

2. Explain why sound is classified as a longitudinal wave rather than a transverse wave.

3. A wave has a frequency of 20 Hz and travels at a speed of 340 m/s. Calculate its wavelength. Show your working.

4. Look at a diagram of a slinky wave. How can you tell if the wave is moving to the right or to the left?

5. If you measure the distance from the centre of a compression to the centre of the adjacent rarefaction, what fraction of the wavelength have you measured?

Extension

Research how ultrasound imaging works in medicine. Explain how the generation and detection of compressions and rarefactions in high-frequency sound waves allow doctors to see inside the human body without using X-rays.
